

F-TALENT CODE 2026

Network Infrastructure in Data Centers

Time Limit: 1 second(s) | Memory Limit: 256 MB

Problem description

A company is planning a communication network for N data centers, indexed from 1 to N . There are M candidate cables available for installation. Each cable connects a pair of data centers and carries a specific construction cost. Cables are assigned unique IDs from 1 through M , reflecting their sequential order in the input..

The company wants to select a set of cables such that:

- All N data centers are connected (directly or indirectly);
- The total construction cost of the selected cables is minimized;
- No cycle is formed in the selected cables.

Some cables are marked as mandatory. Every mandatory cable must be included in the final network, regardless of its cost. If it is impossible to build a valid network that includes all mandatory cables, output IMPOSSIBLE.

Because several cables may share the same construction cost, the following tie-breaking rule is used to guarantee a unique answer: if multiple valid minimum-cost networks exist, the network whose selected cable IDs, sorted in increasing order, form the lexicographically smallest sequence must be chosen.

For two sorted lists of cable IDs, compare their elements from left to right; at the first position where the two lists differ, the list containing the smaller ID is lexicographically smaller. For example, $[1, 3, 5] < [1, 4, 5]$, and $[1, 2, 7] < [1, 3, 4]$.

Your task:

Given N , M , K , the list of M cables (each with two endpoints and a cost), and the list of K mandatory cable IDs, determine:

- whether a valid network satisfying all constraints exists;
if it exists, the minimum total construction cost, the number of selected cables and the set of cable IDs to build, following the tie-breaking rule described above.

Input / Output specification

INPUT	OUTPUT
The first line contains three integers N, M, Ks: - N: the number of data centers - M: the number of possible cables - K: the number of mandatory cables Each of the next M lines describes one cable, in ID order (the i-th line has ID i), containing three integers u, v, w (the two data centers connected by the cable and its construction cost). If $K > 0$, the next line contains K distinct integers id_1 id_2 ... id_K (the IDs of the mandatory cables.)	If it is impossible to construct a valid network containing all mandatory cables, print a single line: IMPOSSIBLE Otherwise, print three lines: - The total construction cost of the selected cables; - The number of selected cables; - The IDs of the selected cables, in increasing order, separated by space.

Constraints

- $2 \leq N \leq 200000$
- $1 \leq M \leq 300000$
- $0 \leq K \leq N-1$
- $1 \leq u, v \leq N, u \neq v$
- $1 \leq w \leq 10^9$ (costs of different cables may be equal)
- Each cable has a unique ID from 1 to M; each mandatory cable ID appears at most once.

Example 1:

INPUT	OUTPUT
4 5 3 1 2 3 2 3 4 3 1 2 3 4 5 1 4 10 1 2 3	IMPOSSIBLE

Example 2:

INPUT	OUTPUT
5 6 1 1 2 4 2 3 4 3 4 1 4 5 2 1 3 4 2 5 6 3	11 4 1 2 3 4